

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

Approach to normal labour

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Learning objectives:

To have knowledge about the following:

- Definition of labour.**
- Stages of labour.**
- How to diagnose labour.**
- How to manage different stages of labour.**
- How to monitor the progress of labour.**

Definition of labour

Labour is the physiological process by which the fetus, placenta, and membranes (product of conception) are expelled from the uterus through the birth canal.

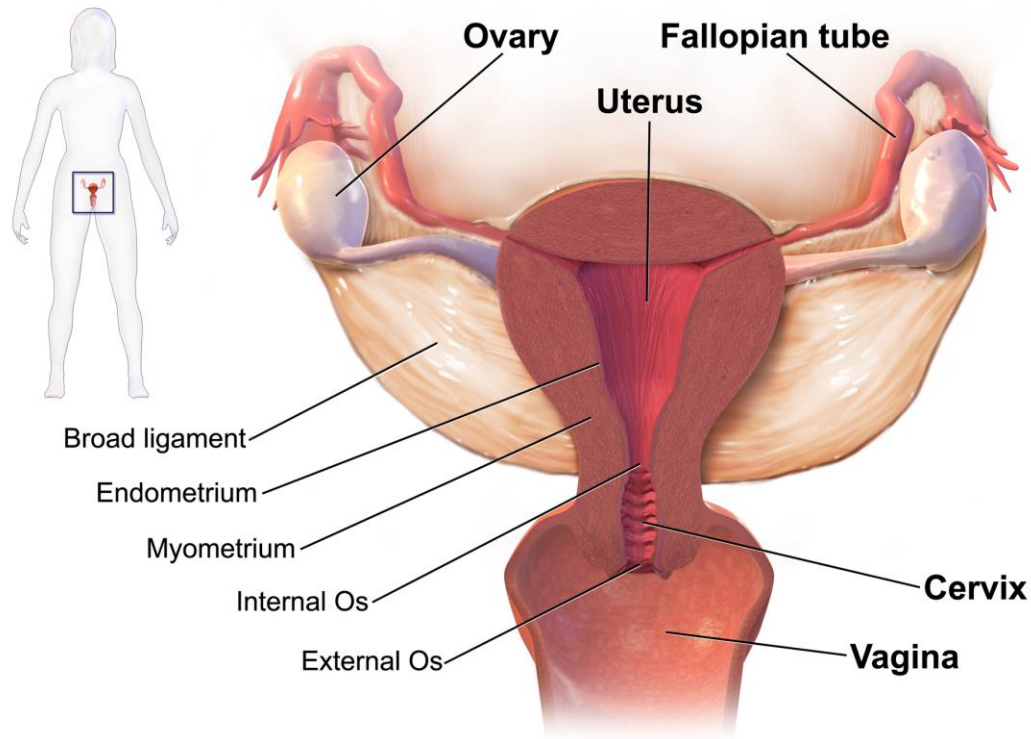
It is characterized by:

1. Regular, painful uterine contractions that increase in frequency, duration, and intensity.
2. Progressive effacement and dilatation of the cervix.
3. Descent of the fetus through the birth canal, leading to delivery.

In short:

Labour = *regular uterine contractions + cervical changes + progressive fetal descent.*

The female reproductive system

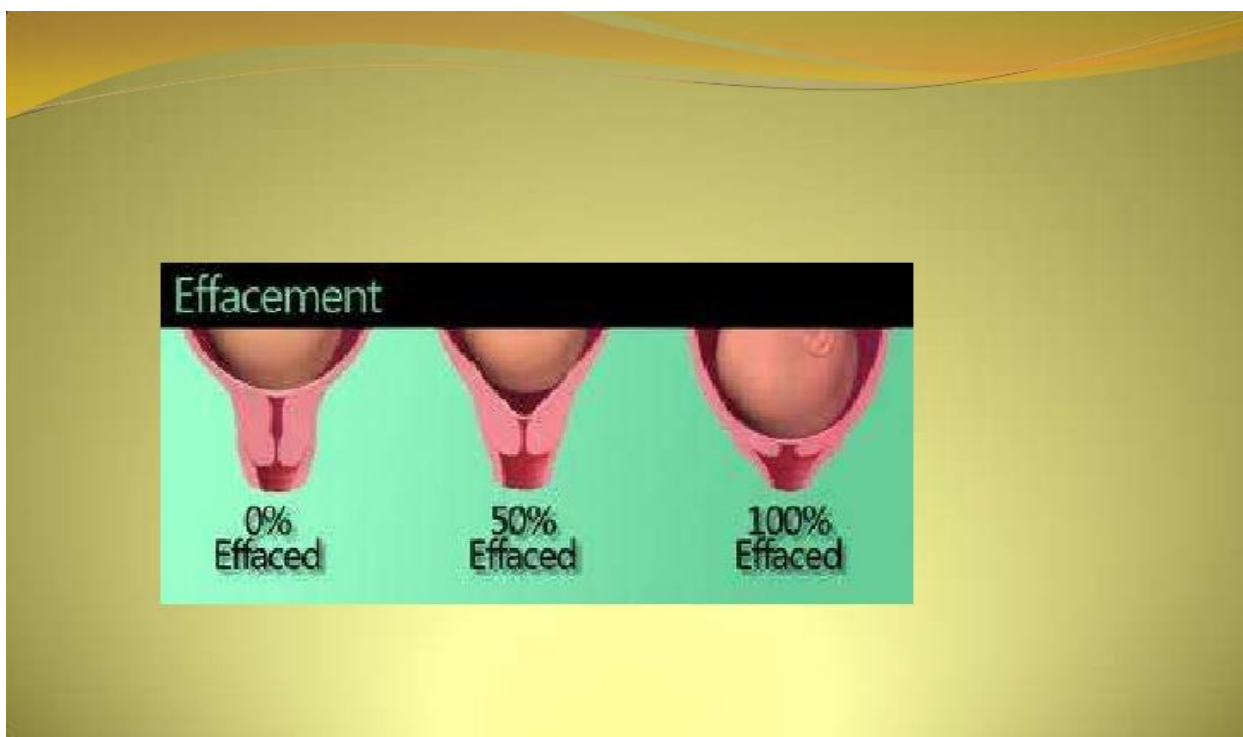


-If expulsion of the product of conception
occurs after 24 weeks of gestation the process
is called labour.

-If expulsion of the product of conception
occurs before 24 weeks of gestation, it is
called spontaneous abortion(miscarriage).

What is the meaning of effacement?

Effacement means progressive thinning and shortening of the cervical canal



Stages of labour:

- **First Stage** – from the onset of true labour pains until full cervical dilatation (10 cm).

- **Latent phase:** slow cervical changes (0–3/4 cm).

- Normal average duration 3-8hr.

- Abnormal: >20hr in primigravida, > 14hr in multiparous.

- Contractions occurs at **least 2 contractions/ 10 minutes** with each **lasting 20-30seconds**.

- contractions are **moderately strong** and are **quite well tolerated without analgesia.**

- **Active phase:** rapid cervical dilatation (from ~4 cm to 10 cm).

- Cervix dilate at least 1cm/hr. in primigravida, 1.5cm/hr. & in multiparous.

- The contractions becomes **more frequent, stronger and more prolonged.** .

- At least **3 contractions / 10 minutes**, with each lasting **≥ 40 seconds**.

Normal duration : **2-6 hours.**

• **Second Stage** – from full cervical dilatation (10 cm) until delivery of the fetus.

1.The passive phase :

-The time between full dilatation and the onset of involuntary expulsive contractions.

-There is no maternal urge to push and the fetal head is still relatively high in the pelvis.

2.The active phase:

- There is a maternal urge to push down with contractions because the fetal head is low, causing a reflex need to bear down with contractions.

- Duration of 2nd stage

- Primigravida: 1-2hr without epidural analgesia.
2-3hr with epidural.

- Multiparous: -half-1hr without epidural.
1-2hr with epidural.

Cervical dilatation



A. Cervix is not effaced or dilated.



B. Cervix is fully effaced and dilated to 1 cm.

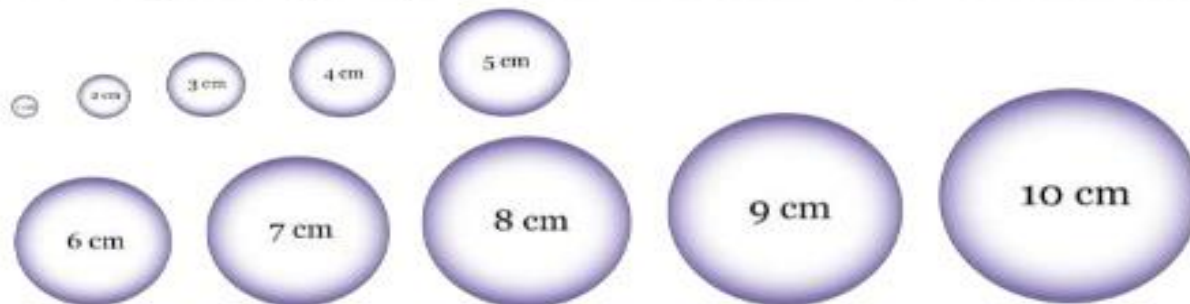


C. Cervix is dilated to 5 cm.



D. Cervix is fully dilated to 10 cm.

Dilation - the gradual opening of the cervix measured in centimeters from 0 to 10 cms.



- **Third Stage** – from delivery of the fetus until **expulsion of the placenta and membranes.**

Duration: 5-15min, abnormal if >30min.

- **Fourth Stage (sometimes described):**

-The first 1–2 hours after delivery, when the mother is observed for postpartum complications (e.g., hemorrhage) is some times called 4th stage.

The Third stage

□ Placenta delivery

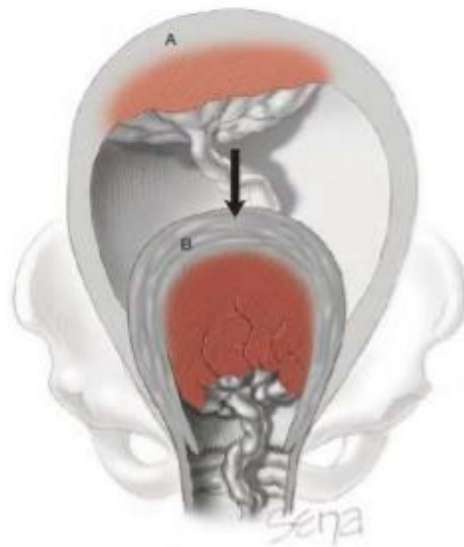
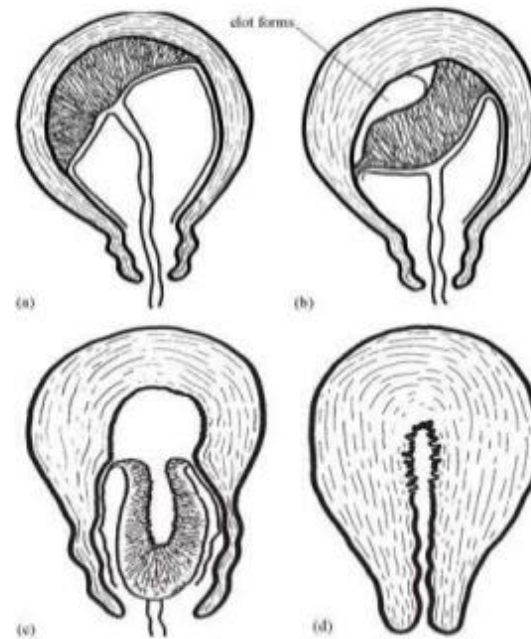


FIGURE 6-10 Diminution in size of the placental site after birth of the infant. **A**, Spatial relations before birth. **B**, Placental spatial relations after birth.



Clinical diagnosis of labour

Symptoms and signs of labour

1. Painful uterine contractions.
2. Shortening and dilatation of the cervix.
3. Passage of cervical mucous (Show).
4. Rupture of membranes .

- Passage of a show (a blood stained plug of mucus passed from the cervix). Or spontaneous rupture of membranes (SRM) does not define the onset of labour.

Uterine contractions:

Throughout pregnancy there are painless irregular uterine contractions called Braxton Hicks contractions.

The contractions of labour characterized by:

1. Comes at regular interval.
2. Contractions are often preceded by backache then spread over the abdomen, it is colicky and tend to increase gradually in frequency, intensity and duration.

The intensity of uterine contractions is assessed by the amplitude of the intrauterine pressure generated with each contraction

-In normal labour this intra-uterine pressure averages between 30 and 60 mmHg (by inserting intrauterine pressure catheter).

- The pain of labour has the same character as that of spasmodic dysmenorrhea and the same cause which is ischemia of the uterine muscles from compression of the blood vessels in the wall of the uterus.
- Labour pain is intermittent(never continuous).
- The intermittent nature of the contractions is of great importance to health of both the fetus and the mother

DIAGNOSIS OF THE ONSET OF LABOUR

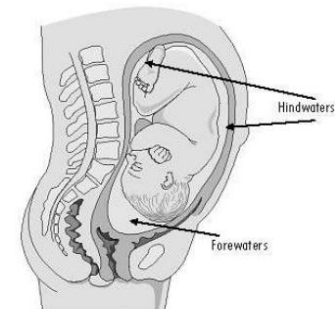
Suspect or anticipate the onset of labour if the woman has:

- Blood stained mucus discharge per vagina (show).
- Watery vaginal discharge.

Confirm the onset of labour if there are the following:

- True regular uterine contractions.
- Progressive cervical effacement.
- Progressive cervical dilatation.

Formation of forewater



The women should be advised to come to hospital when:

- Uterine contractions are occurring every 5–10 minutes.
- Their membranes rupture and amniotic fluid is released.

On admission:

1. Take proper full history.
2. Do Proper examination including:
 - BMI.
 - Vital signs: PR, BP, Temp.
 - Abdominal: lie and presentation of the fetus (breech or cephalic).
 - Degree of engagement of the presenting part (rule of 1/5).
 - Assess uterine contractions.
 - Listen to fetal heart rate.
 - Vaginal examination to assess:
 - Degree of effacement and dilatation of the cervix.
 - The membranes whether intact or ruptured,
 - if ruptured check the amount and color of the fluid.
 - The level of the fetal presenting part in the pelvis.
3. Send urine sample for protein, ketones, glucose and blood.

Management of the 1st stage of labour

What are the things that you should monitor in labour?

1. Care and monitoring of the labouring woman.
2. Monitoring the process of labour.
3. Monitoring the fetal condition.

1. Care of pregnant woman in labour:

1. The woman should **not be left alone** during labour.
 - Ideally there should be a midwife present with her throughout.
 - In addition, one of her relatives.
2. Check vital signs(PR, BP, RR &Temp) **every 2 hours.**
- 3- Mobility: encourage **mobility if the membranes are intact** or they are **ruptured with well engaged head.**
4. Position: allow the women to adopt the position she finds most comfortable. However **avoid the dorsal supine position.**
5. Provide adequate analgesia e.g.
Pethidine (50-150mg IM at intervals of 3-4 hours unless delivery is expected after 1-3 hours)
 - **Entonox** (it is a 50/50 mixture of nitrous oxide with oxygen).
 - **Epidural analgesia** is the gold standard analgesia for labour pain.

6. Food: Light snacks, soup or cool fluids are offered.
- There is no evidence to support the routine insertion of an IV cannula and IV infusion during labour.
 - If administer IV fluids, use Saline or Ringer's lactate, Do not use glucose solutions, especially D10 W.
7. Encourage woman to empty her bladder regularly.
8. Encourage the woman to evacuate the rectum voluntarily.

2. Monitoring the process of labour:

- Check uterine contractions (number and duration) over 10 minutes every half an hour.
- Check for any gush of fluid per vagina (make note of the time and manner of membrane rupture and the color of liquor.)
- Perform vaginal examination to assess :
 - Cervical dilatation and effacement.
 - Descent of the presenting part .
 - Presence of moulding or caput.
- The frequency of vaginal examination should be individualized according to the case (every 3-4 hours in the latent phase and every 1-2 hours in the active phase.
- A vaginal examination should be done after rupture of the membranes to exclude cord prolapse.

-Progress of labour as well as maternal and fetal condition are recorded on a graphic paper called the partogram.

-Partogram: is graphical method of assessing the progress of labour as well as the fetal and maternal condition.

-Partogram helps to identify at an early stage those women whose labour is slow (abnormal progress).

-To recognize maternal or fetal problems as early as possible.

-The partogram used by most maternity units, it is an easy and helps facilitate handover between doctors and midwives.

Component of the partograph:

- Part I : assessment of fetal condition.**
- Part II : progress of labour.**
- Part III : assessment of maternal condition.**

The partogram

Name	Gravida	Para	Hospital number
Date of admission	Time of admission	Ruptured membranes	hours

Fetal heart rate

Amniotic fluid Moulding

Cervix (cm) (Plot X)

Descent of head (Plot O)

Hours

Time

Contractions per 10 min.

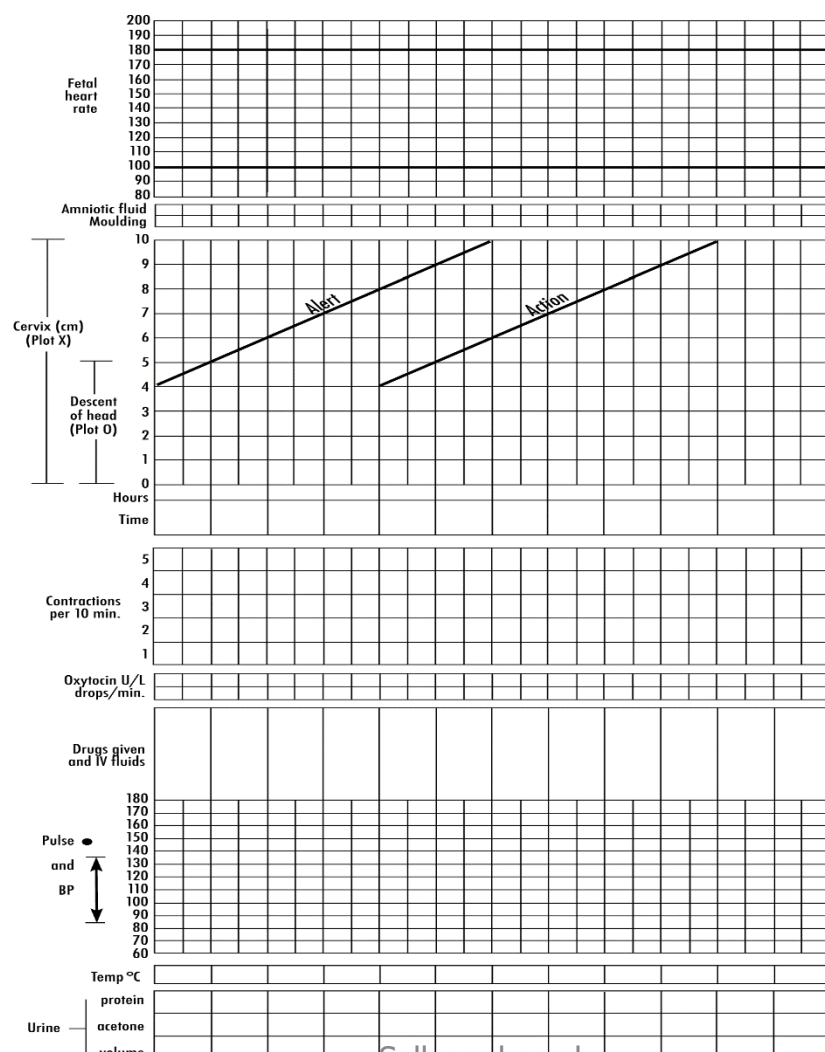
Oxytocin U/L drops/min.

Drugs given and IV fluids

Pulse and BP

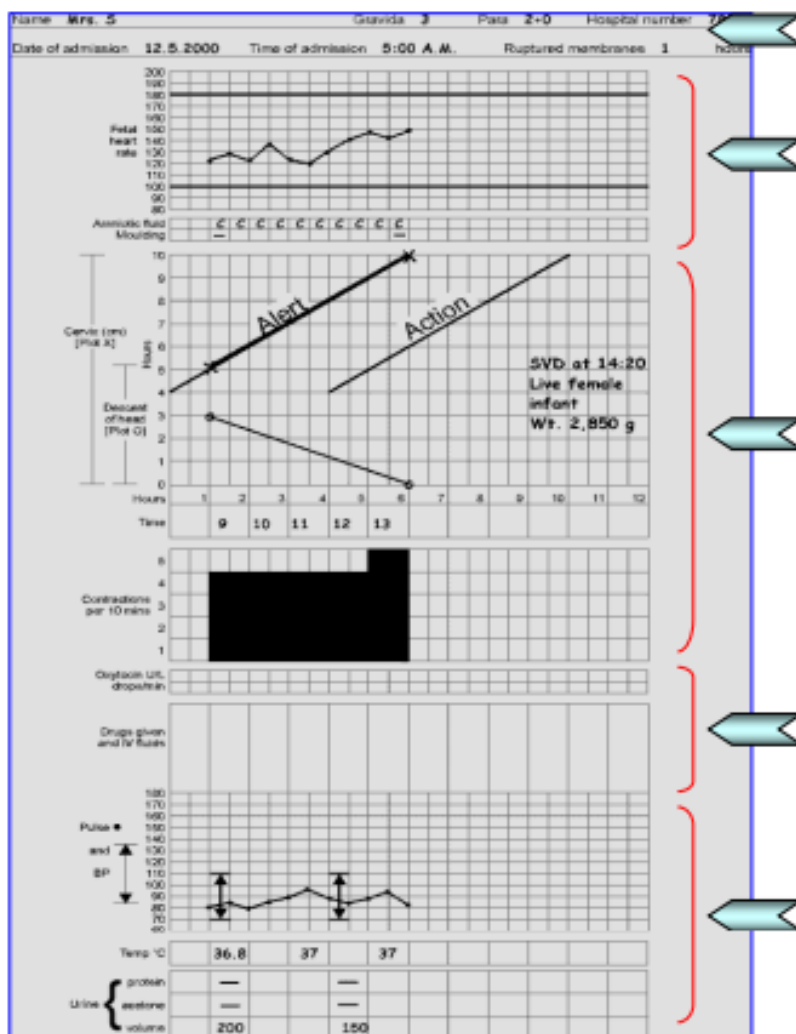
Temp °C

Urine protein acetone volume



The grid shows a vertical axis for time (0-20 hours) and a horizontal axis for various parameters. Two diagonal lines are plotted: the 'Alert' line (from 4 cm at 0h to 10 cm at 10h) and the 'Action' line (from 4 cm at 10h to 10 cm at 20h). The area between these lines is shaded.

PARTOGRAM



Mother information

Fetal well-being

- Fetal heart rate•
- Character of liquor•
- Moulding•

Labour progress

- Dilatation•
- Descent•
- Uterine contraction•

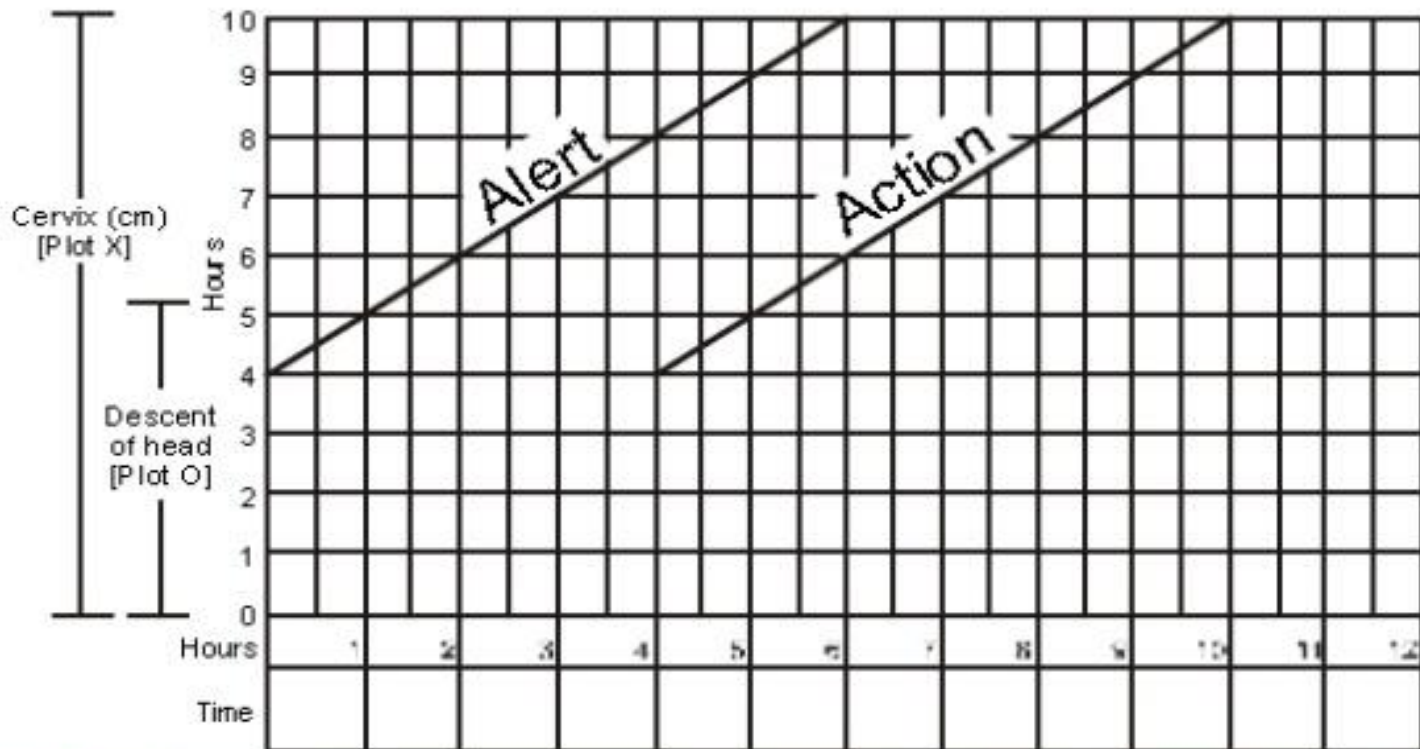
Medications

- Oxytocin•
- Pain relief (e.g. pethidine)•

Maternal well-being

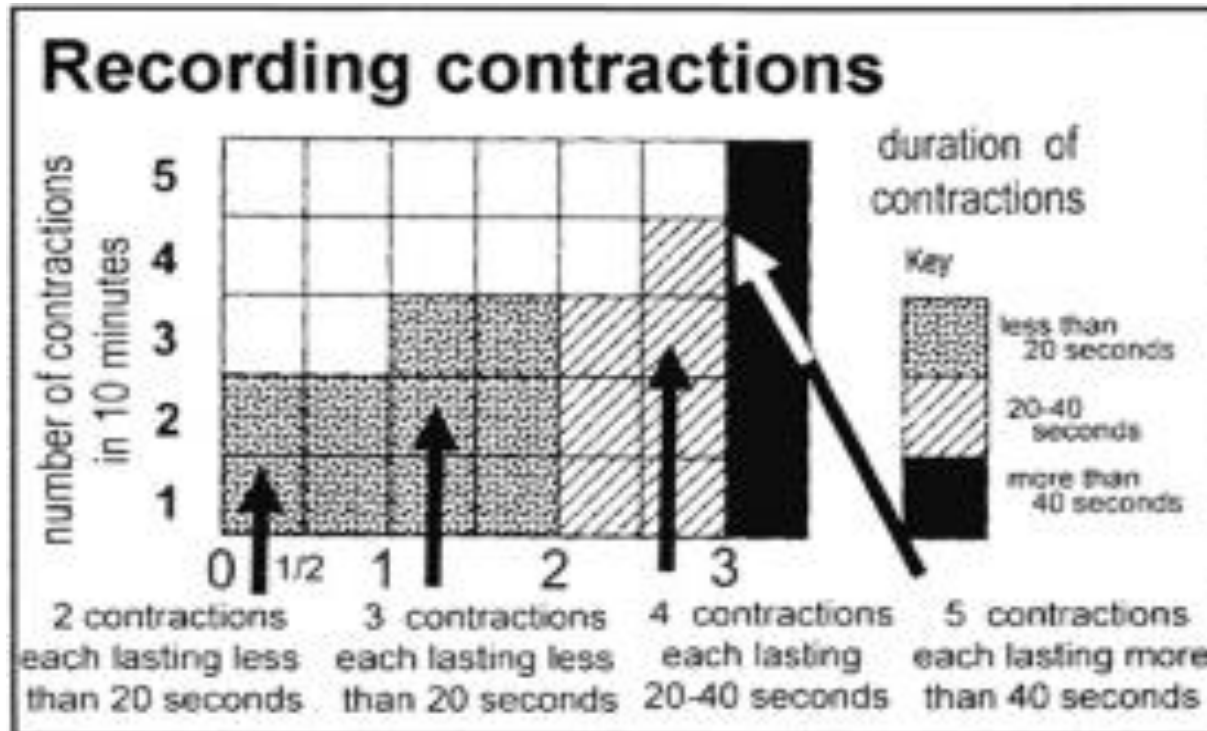
- BP, Pulse, Temperature•
- Urine – albumin, glucose, acetone•
- Urine output•

Cervical dilatation

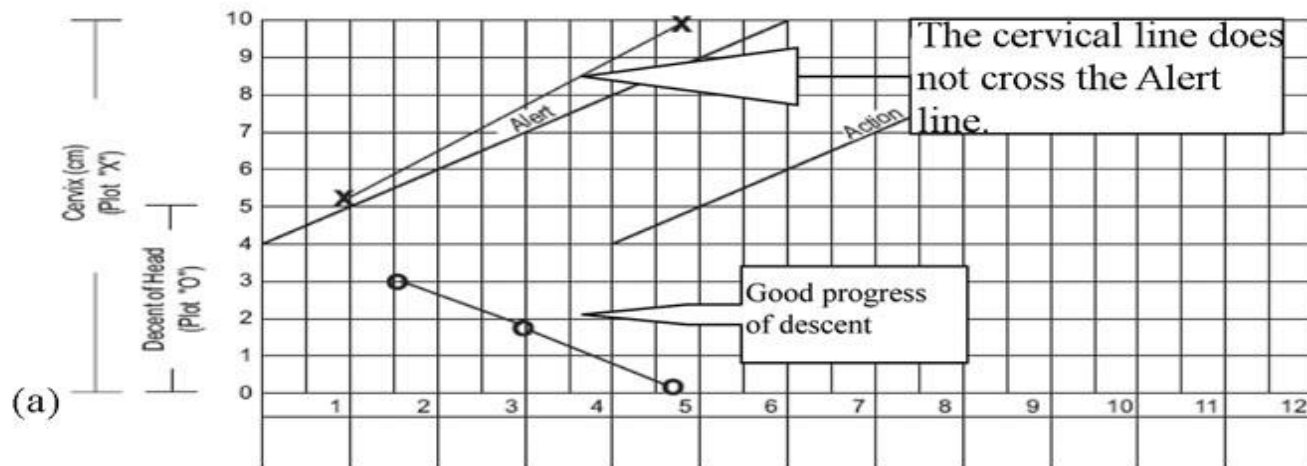


Uterine contractions are assessed every 30minutes for 10min

- 1.The number of squares filled indicate the number of contractions per 10 min.
- 2.The way of filling the square indicate the duration of contraction.



Example of normal progress of labour (a), and abnormal progress(b).



Augmentation of labour:

Definition:

Augmentation of labour is the stimulation of spontaneous uterine contractions that have been considered inefficient because of poor progress in cervical dilatation and /or descent of the presenting part of the fetus.

Methods of augmentation:

1.Amniotomy (artificial rupture of membranes):

-When augmentation becomes necessary the first approach should be to rupture the membranes which found to increase the strength of uterine contractions & shorten the duration of labour .

2. Oxytocin infusion:-

-Intravenous infusion of synthetic oxytocin, usually after either spontaneous or artificial rupture of the membranes, is the most widely used treatment to augment labour when progress is inadequate due to inefficient uterine contractions.

Management of the second stage of labour

- The 2nd stage of labour is the stage of fetal expulsion.
- It begins with full dilatation of the cervix and ends with delivery of the fetus.
- The 2nd stage of labour is a serious stage ; the utero-placental perfusion is markedly reduced due to the strong contractions, resulting in decreased oxygenation of the fetus.
- Therefore the fetus should be carefully monitored during the 2nd stage
- Many factors influence the duration of the 2nd stage of labour:
 - Parity (shorter),
 - Use of Epidural analgesia (longer),
 - Strength of contractions,
 - Fetal size &
 - Perineal resistance.

Preparation for delivery

- Place the labouring woman in the dorsal lithotomy position
- Scrub the vulva and perineum and cover with sterile towels.

Clinical assistance of delivery:

The goals of clinical assistance at delivery are :

- Reduction of maternal perineal trauma.
- Prevention of fetal injury.
- Initial support of the newborn if required.

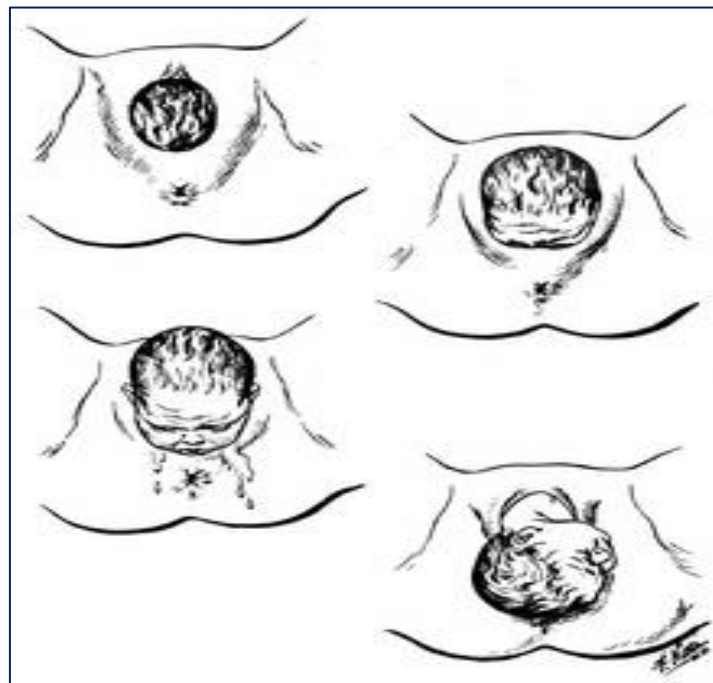
Delivery of the head:

-When the fetal head crowns and delivery is imminent, allow gradual extension of the fetal head by supporting the perineum with one hand and applying downward pressure on the occiput with the other hand to control delivery, thereby preventing rapid expulsion, which has been associated with perineal tears as well as intracranial trauma.

-Crowning mean the biparietal diameter of the fetal head distending the vulval outlet and there is no recession of the head between contractions.

-As the head is delivered, the face is quickly wiped and the fluid in the nose and mouth are aspirated.

Delivery of the head



Management of the third stage of labour:

- The third stage of labour is the stage of separation and expulsion of the placenta.
- It starts immediately after delivery of the neonate and ends with delivery of the placenta and membranes.

Placental separation is aided by:

- Sudden diminution in uterine size due to contraction and retraction of uterine muscles following delivery of the fetus
- Post partum haemorrhage is more likely at this time.

Management of third stage

- 1.Expectant management (old method).**
- 2.Active management (used by most of maternity units .**

Active management of the third stage (AMTSL) :

1. Give oxytocin 10 IU IM with the delivery of the anterior shoulder or after delivery of the neonate.

-If oxytocin is not available give other uterotonics as ergometrine 0.2mg IM, Syntometrine (one ampule) IM, or misoprostol 400-600 ug orally or rectally.

-Oral or rectal misoprostol can be used in situations when safe administration and / or appropriate storage conditions for injectable oxytocin and ergot is not possible.

2. Clamping of the cord 1-3 min after delivery of the fetus (in order to allow some blood to go to the baby from the placenta

-The cord is clamped close to the perineum.

-The cord should be immediately clamped in the following cases:

*Rh incompatibility, Preterm labour, Cases with possible Poly-cythemia of the fetus (maternal DM, IUGR).

**Wait for the signs of placental separation which include
(3signs):**

- 1.Lengthening of umbilical cord.**
- 2.A gush of blood from vagina signifying separation of placenta from uterine wall**
- 3.Change in shape of uterine fundus from discoid to globular with elevation of fundal height.**

Then when the uterus contract, deliver the placenta by:
controlled cord traction

Controlled cord traction

-With the left hand on the abdomen pushing the uterus upward and backward, controlled traction is applied by the right hand on the Cord in a downward direction to deliver the placenta (Brandt Andrews method).

Applying cord traction before placental separation may cause uterine inversion



After delivery the placenta should be checked for any missing piece



Care of the mother immediately after delivery of the placenta (fourth stage of labour).

- The mother should be carefully monitored during the first two hours postpartum:
- Check the general condition of the mother (pulse rate, blood pressure and temperature every half an hour).
- Check if the uterus is contracted (if poorly contracted gentle uterine massage can be helpful).
- Check the uterine Fundal height (blood may accumulate in the uterine cavity).

Important consideration in dealing with labouring woman:

- 1.Reassure labouring women on admission.**
- 2.Speak to the patient with respect and dignity.**
- 3.Explain all the procedures to be done.**
- 4.Keep the patient covered during the examination and when transferring her from one room to another.**
- 5.Respects the woman's pain and her right to have analgesia.**
- 6.Keep the family informed of the progress.**



Thank you